

Course Name: Pharmacognosy

Course code: Phar 2082

Module Name: Pharmacognosy and Alternative Medicine I

Module Code: 08

Course ECTS: 7

Totally required hours for the module: 189 hours

Assessment:

- Seminar/Assignments/ quizzes: 35%
- Laboratory report writing and written exam: 15%
- Practical exam: 10%
- Final Exam: 40%

Year/Semester Course is offered: Year II Semester II

Course prerequisite/s: Chemistry of Natural Products

Course description: The course is designed in such a way that the trainee gets well acquainted with the study of the physical, chemical, biochemical and biological properties of drugs, drug substances, or potential drugs or drug substances of natural origin as well as the search for new drugs from natural sources. The course familiarizes trainees with the basic scientific knowledge and skill needed to obtain and characterize active substances from natural sources. It also helps trainees to understand and realize the fact that nature provides the origin and continuous supply of drugs or drug substances, and think about the proper management and utilization of such natural products.

Course objective:

After completion of this course students will be able to:

- To familiarize themselves to general aspects of crude drugs, extraction and isolation methods and the distribution, properties and uses of various primary and secondary metabolites of plant, animal and mineral origin.

Course mode of delivery: Parallel

Course learning and teaching methods

- Illustrated Lecture: 48 hours
- Practical sessions: 48 hours
- Tutorial: 12 hours
- Seminars, assignments and presentation: 16 hours
- Assessment (continuous & final): 10 hours
- Independent study (alone or in groups): 45 h hours
- Field visits (10 h)

Assessment techniques:

- Seminar/Assignments: 10%
- Quizzes/Continuous assessment: 25%
- Laboratory written exam: 15%
- Practical exam: 10%
- Final Exam: 40%

Teachers' and students' role

Roles of Instructors

- The instructor will be expected to:
- Be a facilitator (introduce the subject; give guidance, moderate discussions, etc.)
- Read and comment assignments of students on time;
- Prepare his/her lessons and deliver lectures;
- Provide available and necessary reference materials;
- Encourage active participation of students in the teaching learning process;
- Assist students with learning difficulties and
- Arrange and follow up practical sessions
- Roles of Students

Students are expected to:

- Engage in learning by doing (independent study, project work; group work, etc.)
- Be active learners (participate effectively in laboratory activities, in group assignments, make presentations, write reports, etc.);

References:

Required readings (Text)

- William C. Evans, George E. Trease, and Daphne Evans, Trease and Evans' Pharmacognosy (16th ed.), Elsevier (2009).

Recommended readings

- Dewick PM. Medicinal Natural Products: A biosynthetic Approach, 3rd edition. John Wiley and Sons, LTD, England 2009.
- Sarker D, Latif Z, Gray A. Methods in Biotechnology Natural Products Isolation, 2nd edition, Human Press, Totowa, New Jersey, 2006
- Pulok K. Mukherejee. Quality control of herbal drugs; an approach to evaluation of botanicals. Business Horizons pharmaceutical publishers 2002

Course schedule*

Week	Content Hours	Topic/sub-topic/chapter/Assessments/Assignments
1&2		- General Introduction - Definition, History and scope of Pharmacognosy (1 hr) - Crude drugs (6 h) - Pharmaceutical botany - Definition and Nomenclature of crude drugs - Classification of crude drugs - Evaluation of crude drugs - Types of preparations from plants - Schemes for pharmacognostic studies of crude drugs. - Official and Unofficial drugs - Factors affecting crude drug quality - Steps in the scientific analysis of drugs from natural resources - Selection of plant material - Taxonomic identification of the plant - Literature survey on the identified plant - Design of appropriate extraction and separation methods - Checking extracts/ fractions for pharmacological activity - Identification of classes of compounds found in the plant

		(phytochemical screening) 1.3.7. Isolation of active compounds or fractions responsible for the pharmacological activity of the plant
3&4	8hrs	2. General methods in studying constituents of crude drugs 2.1. Extraction (3 hrs) 2.1.1. Definition and the need for extraction 2.1.2. Preparation of plant material for solvent extraction 2.1.3. Choice of suitable solvents 2.1.4. Methods of extraction 2.2. Isolation and purification of active constituents (5hrs) 2.2.1. Classical methods of separation 2.2.2. Modern methods/chromatographic methods 2.2.2.1.
5,6,7, 8,9,10 , 11&1 2	29 hrs	3. Major plant constituents and their botanical sources 3.1. Primary and secondary plant metabolites (1 h) 3.2. Carbohydrates (2 h) 3.2.1. Sugars and sugar containing drugs 3.2.2. Compounds related to sugars 3.2.3. Polysaccharides 3.2.4. Gums and mucillages 3.3. Glycosides (6 h) 3.3.1. General properties of glycosides 3.3.2. Classification of glycosides 3.3.3. Classes of glycosides: Anthraquinones, Saponins, Cardiac glycosides, Simplephenolic glycosides, Flavonoid glycosides, Isothiocyanate glycosides, Cyanogenetic glycosides, Coumarin glycosides, lignans 3.4. Tannins (2 h) 3.4.1. General properties and Chemistry 3.4.2. Classification: Hydrolysable, Nonhydrolysable (condensed), Pseudotannins 3.4.3. Significance of tannins

		3.5. Lipids and waxes (2 h) 3.5.1. Lipids: Physical and chemical properties, Extraction methods 3.5.2. Official fixed oils/fats and their composition Arachis oil, castor oil, almond oil, sesame oil, theobroma oil, cod liver oil etc 3.5.3. Waxes: Definition and general properties, Animal waxes, Vegetable waxes 3.6. Volatile oils (5 h) 3.6.1. Distribution and occurrence, Uses, Methods of preparation (Distillation, Expression, Extraction with solvent, enzymatic hydrolysis), Physical properties, Chemistry, Biosynthesis 3.6.2. Constituents of volatile oils: Hydrocarbons, Alcohols, Aldehydes, Ketones, Esters, Phenols and phenolic ethers Oxides Peroxides Sulfur containing compounds Nitrogen containing compounds 3.7. Resins and resin combinations (3 h) 3.7.1. General properties and chemistry 3.7.2. Examples of drugs containing resins: Rosin, Podophyllin, Jalap, Mastic, Cannabis (Preparation, Constituents, Factors affecting the narcotic activity, Legal aspects, Analysis), Oleoresins Oleo-gum resins, Balsams 3.8. Alkaloids (8 h) 3.8.1. Definition, Nomenclature, Occurrence, Physical and chemical properties, Detection, Extraction and isolation 3.8.2. Classification 3.8.3. Classes of alkaloids 3.8.3.1. Ornithine derived alkaloids: Tropane alkaloids Solanaceae alkaloids, Coca alkaloids, Pyrrolizidine alkaloids 3.8.3.2. Lysine derived alkaloids: Lobelia alkaloids, Lupine alkaloids 3.8.3.3. Nicotinic acid derived alkaloids: Ricinine, Areca alkaloids, Tobacco alkaloids 3.8.3.4. Tyrosine derived alkaloids: (1) Simple phenylethylamines and tetrahydroisoquinolines: Mescaline, Ephedrine, Alkaloids of 'Khat'; (2) Colchicine; (3) Papaver
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		alkaloids: Phenanthrene group, Benzyloisoquinoline group; (4) Emetine and related alkaloids; (5) Tubocurarine 3.8.3.5. Tryptophan derived alkaloids: Simple indoles, Tricyclic alkaloids: Pegamum alkaloids, Physostigma alkaloids, Ergot alkaloids (Historical background, Life cycle of ergot, Commercial production, Chemistry and occurrence, Biosynthesis and use, LSD), Rauwolfia alkaloids, Strychnos alkaloids, Cinchona alkaloids, Perwinkle alkaloids 3.8.3.6. Histidine derived alkaloids: Pilocarpine 3.8.3.7. Polyacetate derived alkaloids: Hemlock alkaloids 3.8.3.8. Psedoalkaloids: Steroidal alkaloids: Veratrum alkaloids, Solanum alkaloids, Holarrhena alkaloids, Buxus alkaloids, Purine alkaloids
13&1 4	(4 h)	4. Natural compounds and cancer • Anticancer drugs of natural origin • Natural compounds as carcinogens